Astrophysics Club

Meeting 11: Workstation Activity

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Three stations. Three independent pieces of evidence that 95% of the universe is made of stuff we can't see. Your job at each station: be the astronomer who figures out what's hiding.

Station 1

Galaxy Rotation Curves: The Missing Mass

Useful Constants		
$G = 6.674 \times 10^{-11}\ \mathrm{m^3\,kg^{-1}\,s^{-2}}$	$c = 3.0 \times 10^8$ m/s	$M_\odot = 1.989 \times 10^{30}$ kg
1 kpc $= 3.086 \times 10^{19}$ m	1 Mpc $= 3.086 \times 10^{22}$ m	1 km/s $= 10^3$ m/s

Things orbiting farther from a mass should move *slower* — just like Neptune orbits the Sun more slowly than Earth. The expected orbital velocity based on visible mass is

$$v(r) = \sqrt{\frac{GM(r)}{r}}.$$

But when astronomers tracked stars at the edges of real galaxies, the stars were moving *way too fast*. Something invisible had to be pulling on them.

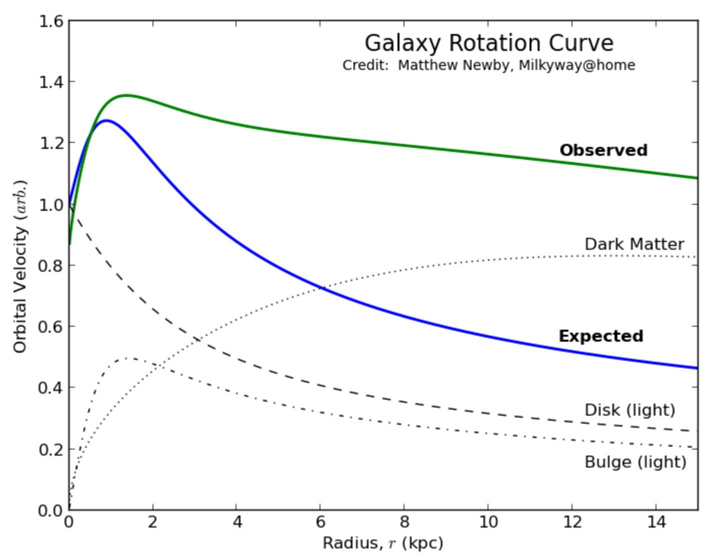

Figure 1: Galaxy rotation curve. The gap between "Expected" and "Observed" is the dark matter problem.

The Mystery of *Pyralis*. You are the astronomer on duty. Your telescope is pointed at the galaxy *Pyralis*, and you need to figure out how much mass is really there.

Before you calculate anything: If *only* visible mass existed, should stars *farther* from the center orbit faster or slower? Circle one. *(Hint: what happens to* $v = \sqrt{GM/r}$ *as* r *grows?)*

FASTER **SLOWER**

(a) Count the visible mass. The galaxy *Pyralis* has a radius of 20.57 kpc. It has roughly 2.3×10^8 stars/(kpc)^2 distributed uniformly across its disk, with each star averaging 1.2 $M_\odot$.

Calculate the total enclosed *visible* mass within a radius of 8.58 kpc. Give your answer in solar masses, scientific notation, two decimal places. *(Hint: what's the area of a disk with radius* 8.58 kpc*?)*

(b) The data comes in. Stars at $r = 8.58$ kpc are actually orbiting at $v = 3.4 \times 10^5$ m/s. Using the rearranged formula

$$M(r) = \frac{v(r)^2\, r}{G},$$

calculate the *actual* enclosed mass within 8.58 kpc. Give your answer in solar masses, scientific notation, two decimal places.

(c) Find the missing piece. Compare your answers from (a) and (b).

- How much "missing" mass is there? _____________ $M_\odot$
- About how many times *heavier* is the actual galaxy than its visible parts? _____________
- What percentage of the total mass is *dark matter*? _____________

Station 2

Gravitational Lensing: Weighing the Invisible

Useful Constants		
$G = 6.674 \times 10^{-11}\ \mathrm{m^3\,kg^{-1}\,s^{-2}}$	$c = 3.0 \times 10^{8}$ m/s	$M_\odot = 1.989 \times 10^{30}$ kg
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Einstein's general relativity predicts that massive objects bend the path of light passing near them. The bigger the mass, the bigger the bend. The deflection angle is

$$\theta = \frac{4GM}{bc^2},$$

where M is the mass of the lens and b is the impact parameter (how close the light passes to the center of the mass).

When astronomers pointed Hubble at galaxy clusters, the light from background galaxies was bending *far* more than visible matter could explain. Something massive and invisible was in the way.

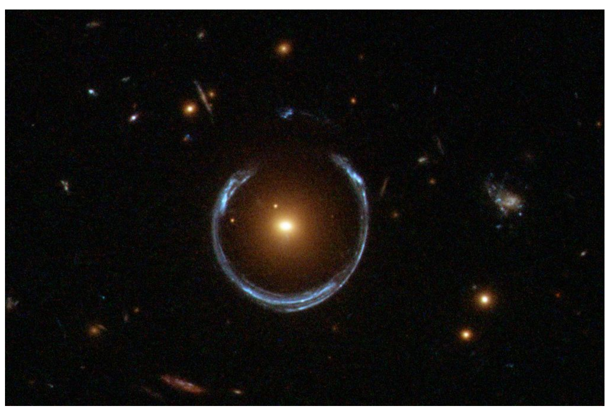

Figure 2: Gravitational lensing around a galaxy cluster.

Bending Light Around *Thalor*. Your telescope has detected the distant galaxy *Thalor*, sitting directly behind an invisible dark matter halo. The halo is bending Thalor's light. From just three numbers, you can weigh the invisible.

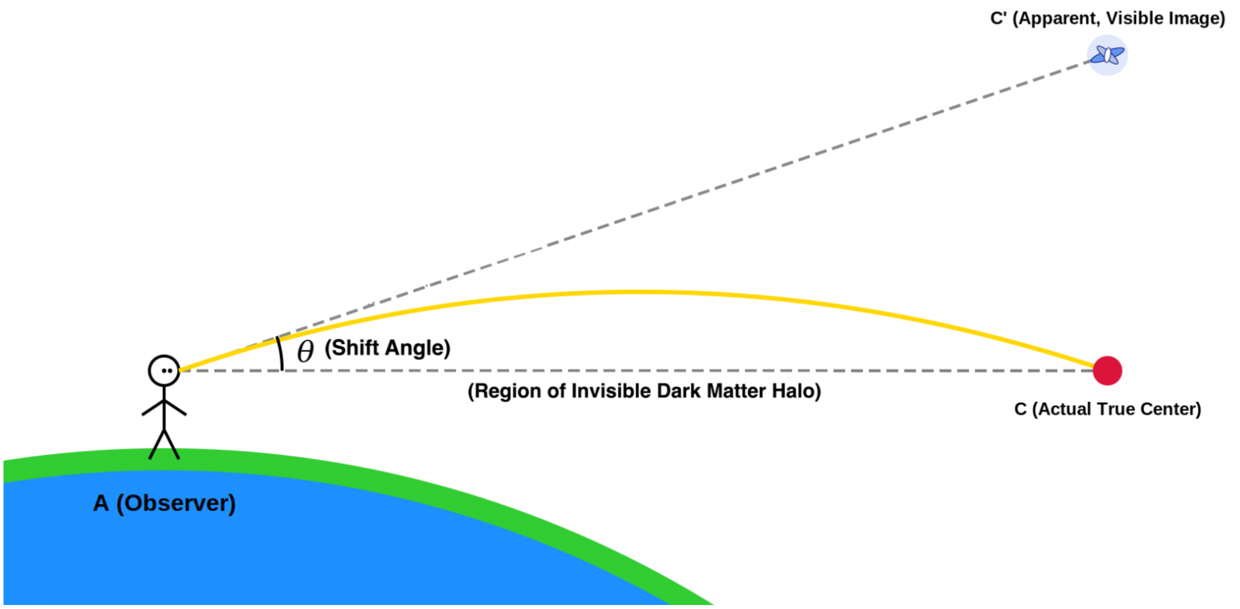

Figure 3: Gravitational lensing by an invisible dark matter halo.

(a) Solve for what you can't see. Rearrange $\theta = \dfrac{4GM}{bc^2}$ to solve for M:

$$M = \underline{\hspace{6cm}}$$

(b) Find the angle. For small angles (as always in lensing), we can use the small-angle approximation:

$$\theta \approx \frac{\text{shift}}{\text{distance}}.$$

Thalor's image appears shifted by 0.0068 Mpc over a true distance of 4.789 Mpc. Calculate θ in radians.

(c) Weigh the invisible. The impact parameter is $b = 0.232$ Mpc. Plug into your formula from (a) and calculate the mass of the dark matter halo. Give your answer in solar masses, scientific notation, two decimal places.

Station 3

The Fate of Valdris

Useful Constants		
$G = 6.674 \times 10^{-11}\ \mathrm{m^3\,kg^{-1}\,s^{-2}}$	$c = 3.0 \times 10^8$ m/s	$M_\odot = 1.989 \times 10^{30}$ kg
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The Friedmann equation is the master equation of cosmology. It links the expansion rate of the universe to everything inside it:

$$H^2 = \frac{8\pi G}{3}\rho - \frac{kc^2}{a^2} + \frac{\Lambda c^2}{3}.$$

There's a single density at which a universe is *just barely* flat — the **critical density**:

$$\rho_{\mathrm{crit}} = \frac{3H_0^2}{8\pi G}.$$

Every density in the universe is measured as a fraction of ρ_{crit}, called a *density parameter*:

$$\Omega_m = \frac{\rho_m}{\rho_{\mathrm{crit}}}, \qquad \Omega_\Lambda = \frac{\rho_\Lambda}{\rho_{\mathrm{crit}}}, \qquad \Omega_m + \Omega_\Lambda + \Omega_k = 1.$$

The sign of Ω_k is the fate of the universe:

$\Omega_k = 0$ **flat** — expands forever, decelerating to zero
$\Omega_k > 0$ **open** — expands forever
$\Omega_k < 0$ **closed** — gravity wins, universe eventually recollapses

Problem. A team of astronomers announces the discovery of a new universe they call *Valdris*. From years of observations, they report the following:

- Hubble parameter: $H_0 = 70.0$ km/s/Mpc
- Measured matter density: $\rho_m = 2.76 \times 10^{-27}\ \mathrm{kg/m^3}$
- Dark energy density parameter: $\Omega_\Lambda = 0.65$

Your job is to decide the fate of Valdris.

(a) Compute Valdris's critical density ρ_{crit}. First convert H_0 to SI units ($\mathrm{s^{-1}}$), then evaluate the formula. Give your answer in $\mathrm{kg/m^3}$, scientific notation, two decimal places.

(b) Compute Valdris's matter density parameter Ω_m. Round to two decimal places.

(c) Use $\Omega_m + \Omega_\Lambda + \Omega_k = 1$ to solve for Ω_k. Based on its sign, is Valdris **flat**, **open**, or **closed**? In one sentence, what is the ultimate fate of Valdris?

(d) Further observations reveal that 85% of Valdris's matter is dark matter and only 15% is ordinary (baryonic) matter. **What fraction of Valdris's total energy budget is ordinary matter?** In our own universe, ordinary matter makes up roughly 5% of the total energy budget — how does Valdris compare? Is it more, less, or about the same?